Exam Date & Time: 31-Dec-2024 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.Tech. (ELECTRONICS & COMMUNICATION ENGINEERING) DEGREE EXAMINATIONS - DECEMBER 2024 / JANUARY 2025
SUBJECT: ECE 2123/ECE_2123 - SIGNALS & SYSTEMS
(MAKEUP)

SIGNALS and SYSTEMS [ECE 2123]

Marks: 50 **Duration: 180 mins.**

A

Answer all the questions.

1A) Use the method of partial fractions to find the time domain signal whose unilateral Laplace transform is given by

$$X(s) = \frac{4s^3 + 3s^2 - 20s + 11}{s^2 + s - 6}$$ (4)

1B) A trapezoidal signal x(t) is defined as given below.
Sketch (i) x(4t − 2)x(t + 4), (ii) x(0.5t) + x(t)

$$x(t) = \begin{cases} 4 + t, & -4 \le t \le -3 \\ 1, & -3 \le t \le 3 \\ 4 - t, & 3 \le t \le 4 \\ 0, & otherwise \end{cases}$$ (3)

1C) Determine the FS representation of the signal shown in Fig. 1C.

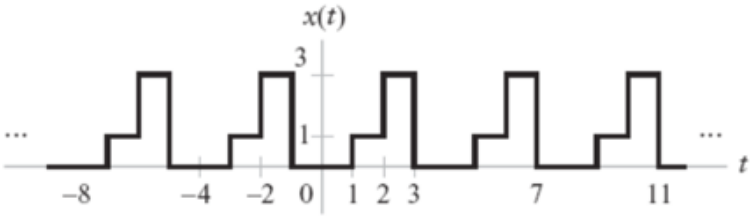

Fig. 1C

(3)

2A) Use the following clues to determine the impulse response h[n] and the transfer function H(z):
H(z) has poles at z = 1/3 and z = −1, h[1] = 1/2, h[−1] = 2 and ROC includes the point z = 3/4. (4)

2B) A rectangular pulse is defined by

$$x(t) = \begin{cases} A, & 0 \le t \le T \\ 0, & Otherwise \end{cases}$$

It is applied to an integrator defined by

$$y(t) = \int_0^t x(\tau)d\tau$$ (3)

Determine the energy spectral density of (i) x(t) and (ii) y(t).

2C) Evaluate the convolution $y(t) = 2t^2\{u(t+1) - u(t-1)\} * 2u(t+2)$ (3)

3A) Certain discrete time LTI system has impulse response $h[n] = \frac{1}{4}\delta[n] + \left(\frac{1}{4}\right)^n u[n] - 2\left(\frac{1}{4}\right)^n u[n-1]$. Write the difference equation representing this system. Use frequency domain analysis to determine the input required to produce an output of $y[n] = \left(\frac{1}{4}\right)^n u[n] + \frac{1}{4}\left(\frac{1}{2}\right)^n u[n]$ for this system. (4)

3B) Certain continuous time LTI system has impulse response $h(t) = \frac{\sin\left(\frac{5\pi t}{2}\right)}{\pi t}$. Use FT to determine the system output if the input is periodic rectangular wave with period T=2 defined by (3)

$$x(t) = \begin{matrix} 1, for\ |t| \le 0.5 \\ 0, for\ 0.5 < |t| \le T/2 \end{matrix}$$

3C) Consider two discrete time signals

$$x[n] = \begin{matrix} 1, \ for\ 0 \le n \le 3 \\ -1, \ for\ 4 \le n \le 7 \\ 0, \ elsewhere. \end{matrix} \quad \text{and}$$

(3)

$$h[n] = \left(\frac{1}{2}\right)^n u[n].$$

Let y[n] = x[n]*h[n] where * denote convolution sum.
Determine y[3], y[5] and y[8].

4A) Consider the discrete time system depicted in the figure Q4A.
Let $h[n] = \frac{\sin(\frac{n\pi}{2})}{n\pi}$. Determine the output response $y[n]$ for the input, $x[n] = \delta[n] - \frac{\sin(\frac{n\pi}{2})}{n\pi}$, and $w[n] = (-1)^n$
Also sketch $G(e^{j\Omega})$ and $Y(e^{j\Omega})$. (4)

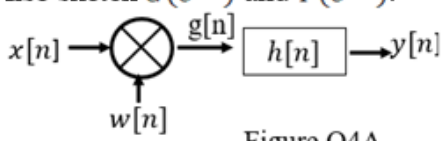

Figure Q4A

4B) Sketch the signal $w[n] = x[3-n] \cdot y[-2-n]$ if
$x[n] = \{3,1,2,\underset{\uparrow}{1},2,1,3\}$ and
$y[n] = \{-1,-1,-1,-1,\ \underset{\uparrow}{0},\ 1,\ 1,\ 1,\ 1\}$ (3)
Also sketch the even and odd components of $w[n]$.

4C) Determine and sketch the power spectral density (PSD) of the signal $x[n] = 1 + \cos\left(\frac{10(n-3)\pi}{19}\right) + 2\sin\left(\frac{14(n-3)\pi}{19}\right)$. (3)
Determine the average power of $x[n]$ using its PSD.

5A) (4)

Consider sampling the signal:
$x(t) = cos(\omega_0 t)$,
with a sampling interval of Ts = 1/4 seconds. For each of the following cases, plot the spectrum of the sampled signal and identify whether aliasing occurs:

a. $\omega_0 = 6\pi$ radians/second

b. $\omega_0 = 4\pi$ radians/second

c. $\omega_0 = 2\pi$ radians/second

5B) Using properties, find the Fourier representation of

$$x[n] = 2jn\, e^{j\frac{\pi}{6}n}, |n| \leq 4$$

(3)

5C) Given the transfer function:

$$H(z) = \frac{1+z^{-1}}{1-\frac{3}{4}z^{-1}+\frac{1}{8}z^{-2}}$$

(3)

Draw the pole-zero plot of the transfer function and specify the region of convergence to ensure the system satisfies both causality and stability conditions Hence, determine the impulse response of the system.

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